

Advanced Numerical Analysis: Iterative Methods, Convergence Analysis, and Complete Problem Solutions

Lecture Companion & Comprehensive Reference Notes

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Abstract

This document provides a self-contained, mathematically rigorous exposition of iterative root-finding techniques in numerical analysis, focusing on condition of convergence, order (rate) of convergence, and optimal parameter design. It covers: (1) General convergence criterion for the Newton-Raphson scheme; (2) Derivation of cubic convergence for square-root iteration; (3) Complete analytical solution of the undetermined parameter problem for evaluating $\sqrt{a}$; (4) Complete analytical solution of the undetermined parameter problem for evaluating $a^{1/3}$; (5) Systematic derivation of Chebyshev-type second-order correction methods; (6) Complete solution and parameter analysis for the non-linear fixed-point scheme $x_{k+1} = \frac{1}{\alpha+1} [\alpha x_k + x_k^{-2} + 1]$; and (7) Full determination of the optimal integer parameter k for the polynomial $x^3 - 5x^2 + 4x - 3 = 0$. All derivations are provided with complete error analyses, asymptotic constants, and numerical verifications.

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1 Theoretical Foundations: Order and Rate of Convergence

Definition 1.1 (Iterative Sequence and Error). Let $f(x) = 0$ be a non-linear algebraic or transcendental equation with an isolated exact root $x = \alpha$, so that $f(\alpha) = 0$. An iterative root-finding algorithm generates a sequence $\{x_k\}_{k=0}^{\infty}$ defined by a continuous

iteration function $\phi(x)$:

$$x_{k+1} = \phi(x_k), \quad k = 0, 1, 2, \dots \quad (1.1)$$

The fixed point of ϕ corresponds to the exact root, i.e., $\alpha = \phi(\alpha)$. The absolute error at step k is defined by:

$$e_k = x_k - \alpha \iff x_k = \alpha + e_k \quad (1.2)$$

Definition 1.2 (Order of Convergence and Asymptotic Error Constant). An iterative sequence $\{x_k\}$ is said to converge to the root α with **order of convergence** $p \geq 1$ if there exists a non-zero finite constant $C > 0$ such that:

$$\lim_{k \rightarrow \infty} \frac{|e_{k+1}|}{|e_k|^p} = C \quad (1.3)$$

where C is termed the **asymptotic error constant**.

- If $p = 1$ and $C < 1$, the convergence is **linear**.
- If $p = 2$, the convergence is **quadratic** (the number of accurate decimal digits roughly doubles at each iteration).
- If $p = 3$, the convergence is **cubic** (accurate decimal digits roughly triple at each iteration).

Theorem 1.1 (Taylor Series Criterion for Order of Convergence). Let $\phi(x) \in C^p[a, b]$ in a neighborhood of the root α . Then the sequence $x_{k+1} = \phi(x_k)$ has order of convergence p if and only if:

$$\phi'(\alpha) = \phi''(\alpha) = \dots = \phi^{(p-1)}(\alpha) = 0, \quad \text{and} \quad \phi^{(p)}(\alpha) \neq 0 \quad (1.4)$$

In this case, the asymptotic error equation is given by Taylor expansion:

$$e_{k+1} = \frac{\phi^{(p)}(\alpha)}{p!} e_k^p + O(e_k^{p+1}) \implies C = \frac{|\phi^{(p)}(\alpha)|}{p!} \quad (1.5)$$

Proof. By Taylor's theorem expanded around $x = \alpha$:

$$\begin{aligned} x_{k+1} &= \phi(x_k) = \phi(\alpha + e_k) \\ &= \phi(\alpha) + e_k \phi'(\alpha) + \frac{e_k^2}{2!} \phi''(\alpha) + \dots + \frac{e_k^p}{p!} \phi^{(p)}(\alpha) + O(e_k^{p+1}) \end{aligned} \quad (1.6)$$

Since $\alpha = \phi(\alpha)$ and $e_{k+1} = x_{k+1} - \alpha$:

$$e_{k+1} = \phi'(\alpha) e_k + \frac{\phi''(\alpha)}{2!} e_k^2 + \dots + \frac{\phi^{(p)}(\alpha)}{p!} e_k^p + O(e_k^{p+1}) \quad (1.7)$$

If $\phi'(\alpha) = \dots = \phi^{(p-1)}(\alpha) = 0$ and $\phi^{(p)}(\alpha) \neq 0$, dividing by e_k^p and taking $k \rightarrow \infty$ gives $\lim_{k \rightarrow \infty} \frac{e_{k+1}}{e_k^p} = \frac{\phi^{(p)}(\alpha)}{p!}$, completing the proof. $\square$

2 Condition of Convergence of Newton-Raphson Method

Problem 2.1 (Notes Page 1). Find the condition under which the Newton-Raphson method converges for finding the root of $f(x) = 0$ in the interval $[a, b]$.

Derivation and Solution: Let the root α of $f(x) = 0$ lie in $[a, b]$. The classical Newton-Raphson iteration formula is:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \quad (2.1)$$

provided $f'(x_k) \neq 0$. This can be viewed as a fixed-point iteration:

$$x_{k+1} = \phi(x_k) \quad \text{where} \quad \phi(x) = x - \frac{f(x)}{f'(x)} \quad (2.2)$$

According to the Contraction Mapping Principle, the iteration $x_{k+1} = \phi(x_k)$ converges to the root α if:

$$|\phi'(x)| < 1 \quad \text{for all } x \text{ in an open neighborhood } I \text{ containing } \alpha \quad (2.1)$$

Differentiating $\phi(x)$ with respect to x using the quotient rule:

$$\begin{aligned} \phi'(x) &= \frac{d}{dx} \left[x - \frac{f(x)}{f'(x)} \right] \\ &= 1 - \frac{f'(x) \cdot f'(x) - f(x) \cdot f''(x)}{[f'(x)]^2} \\ &= 1 - \frac{[f'(x)]^2}{[f'(x)]^2} + \frac{f(x)f''(x)}{[f'(x)]^2} \\ &= \frac{f(x)f''(x)}{[f'(x)]^2} \end{aligned} \quad (2.2)$$

Applying the convergence criterion $|\phi'(x)| < 1$:

$$\left| \frac{f(x)f''(x)}{[f'(x)]^2} \right| < 1 \iff |\mathbf{f}(\mathbf{x})\mathbf{f}''(\mathbf{x})| < [\mathbf{f}'(\mathbf{x})]^2 \quad (2.3)$$

Remarks on Newton-Raphson Convergence:

- Quadratic Nature at the Root:** At the exact root $x = \alpha$, $f(\alpha) = 0$. Assuming $f'(\alpha) \neq 0$, we have:

$$\phi'(\alpha) = \frac{f(\alpha)f''(\alpha)}{[f'(\alpha)]^2} = 0 \quad (2.4)$$

By Theorem 1.1, since $\phi'(\alpha) = 0$, the Newton-Raphson method possesses **at least quadratic convergence** ($p = 2$).

- Asymptotic Error Constant:** Evaluating the second derivative at α :

$$\phi''(x) = \frac{f'(x)f''(x) + f(x)f'''(x)}{[f'(x)]^2} - 2\frac{f(x)[f''(x)]^2}{[f'(x)]^3} \implies \phi''(\alpha) = \frac{f''(\alpha)}{f'(\alpha)} \quad (2.5)$$

Hence:

$$e_{k+1} = \frac{f''(\alpha)}{2f'(\alpha)} e_k^2 + O(e_k^3) \implies C = \left| \frac{f''(\alpha)}{2f'(\alpha)} \right| \quad (2.6)$$

3. Fourier's Sufficient Condition: For monotone convergence without limit cycles or oscillations, choose the initial guess x_0 such that $f(x_0)$ and $f''(x_0)$ share the same algebraic sign:

$$f(x_0) \cdot f''(x_0) > 0 \quad (2.7)$$

3 Rate of Convergence of the Square-Root Formula

Problem 3.1 (Notes Pages 1–2). Find the rate of convergence of the iteration formula:

$$x_{k+1} = \frac{1}{8} x_k \left(6 + \frac{3a}{x_k^2} - \frac{x_k^2}{a} \right) \quad (3.1)$$

designed for evaluating the square root $\sqrt{a}$.

Step-by-Step Derivation: Let $\alpha = \sqrt{a}$ denote the exact root, so that $\alpha^2 = a$. Let $e_k = x_k - \alpha$ and $e_{k+1} = x_{k+1} - \alpha$. Expressing x_k in terms of the error:

$$x_k = \alpha + e_k = \alpha \left(1 + \frac{e_k}{\alpha} \right) \quad (3.2)$$

Substitute into the given iterative scheme:

$$\alpha + e_{k+1} = \frac{1}{8} (\alpha + e_k) \left[6 + \frac{3a}{(\alpha + e_k)^2} - \frac{(\alpha + e_k)^2}{a} \right] \quad (3.3)$$

Since $a = \alpha^2$, we simplify each term individually:

$$\frac{3a}{(\alpha + e_k)^2} = \frac{3\alpha^2}{\alpha^2 \left(1 + \frac{e_k}{\alpha} \right)^2} = 3 \left(1 + \frac{e_k}{\alpha} \right)^{-2} \quad (3.4)$$

$$\frac{(\alpha + e_k)^2}{a} = \frac{\alpha^2 \left(1 + \frac{e_k}{\alpha} \right)^2}{\alpha^2} = \left(1 + \frac{e_k}{\alpha} \right)^2 \quad (3.5)$$

Expanding using the generalized binomial theorem (for $|e_k/\alpha| < 1$):

$$\left(1 + \frac{e_k}{\alpha} \right)^{-2} = 1 - 2 \left(\frac{e_k}{\alpha} \right) + 3 \left(\frac{e_k}{\alpha} \right)^2 - 4 \left(\frac{e_k}{\alpha} \right)^3 + 5 \left(\frac{e_k}{\alpha} \right)^4 - \dots \quad (3.6)$$

$$\left(1 + \frac{e_k}{\alpha} \right)^2 = 1 + 2 \left(\frac{e_k}{\alpha} \right) + \left(\frac{e_k}{\alpha} \right)^2 \quad (3.7)$$

Substitute these expansions into the bracketed expression B :

$$\begin{aligned}
 B &= 6 + 3 \left[1 - 2\frac{e_k}{\alpha} + 3\frac{e_k^2}{\alpha^2} - 4\frac{e_k^3}{\alpha^3} + \dots \right] - \left[1 + 2\frac{e_k}{\alpha} + \frac{e_k^2}{\alpha^2} \right] \\
 &= [6 + 3 - 1] + [-6 - 2]\frac{e_k}{\alpha} + [9 - 1]\frac{e_k^2}{\alpha^2} - 12\frac{e_k^3}{\alpha^3} + O(e_k^4) \\
 &= 8 - 8\frac{e_k}{\alpha} + 8\frac{e_k^2}{\alpha^2} - 12\frac{e_k^3}{\alpha^3} + O(e_k^4)
 \end{aligned} \tag{3.8}$$

Multiplying by $\frac{1}{8}$:

$$\frac{1}{8}B = 1 - \frac{e_k}{\alpha} + \frac{e_k^2}{\alpha^2} - \frac{3}{2}\frac{e_k^3}{\alpha^3} + O(e_k^4) \tag{3.9}$$

Now multiply by $(\alpha + e_k)$:

$$\begin{aligned}
 \alpha + e_{k+1} &= (\alpha + e_k) \left[1 - \frac{e_k}{\alpha} + \frac{e_k^2}{\alpha^2} - \frac{3}{2}\frac{e_k^3}{\alpha^3} + O(e_k^4) \right] \\
 &= \alpha \left[1 - \frac{e_k}{\alpha} + \frac{e_k^2}{\alpha^2} - \frac{3}{2}\frac{e_k^3}{\alpha^3} \right] + e_k \left[1 - \frac{e_k}{\alpha} + \frac{e_k^2}{\alpha^2} \right] + O(e_k^4) \\
 &= \alpha + \left(-\frac{\alpha}{\alpha} + 1 \right) e_k + \left(\frac{\alpha}{\alpha^2} - \frac{1}{\alpha} \right) e_k^2 + \left(-\frac{3\alpha}{2\alpha^3} + \frac{1}{\alpha^2} \right) e_k^3 + O(e_k^4) \\
 &= \alpha + 0 \cdot e_k + 0 \cdot e_k^2 + \left(-\frac{3}{2\alpha^2} + \frac{1}{\alpha^2} \right) e_k^3 + O(e_k^4) \\
 &= \alpha - \frac{1}{2\alpha^2} e_k^3 + O(e_k^4)
 \end{aligned} \tag{3.10}$$

Subtracting α from both sides yields the final error equation:

$$\mathbf{e}_{k+1} = -\frac{1}{2\alpha} \mathbf{e}_k^3 + \mathbf{O}(\mathbf{e}_k^4) \tag{3.11}$$

Conclusion: Since $e_{k+1} \approx C e_k^p$ with $p = 3$ and non-zero constant $C = \frac{1}{2\alpha}$, the formula has **cubic convergence (order 3)**.

4 Unsolved Question 1: Optimal Parameters for Evaluating $\sqrt{a}$

Problem 4.1 (Unsolved in Notes Page 2). The iterative formula:

$$x_{k+1} = x_k \left(p + \frac{qa}{x_k^2} + \frac{rx_k^2}{a} \right) \tag{4.1}$$

is used for evaluating $\sqrt{a}$. Find the values of p, q, r so that it converges with the **maximum rate of convergence**.

Complete Analytical Solution: Let $\alpha = \sqrt{a}$, so $\alpha^2 = a$. Define the relative error $\epsilon_k = \frac{e_k}{\alpha} = \frac{x_k - \alpha}{\alpha}$, so $x_k = \alpha(1 + \epsilon_k)$. Substituting into the formula:

$$x_{k+1} = \alpha(1 + \epsilon_k) \left[p + q \frac{\alpha^2}{\alpha^2(1 + \epsilon_k)^2} + r \frac{\alpha^2(1 + \epsilon_k)^2}{\alpha^2} \right] \tag{4.2}$$

Dividing both sides by α :

$$1 + \epsilon_{k+1} = (1 + \epsilon_k) \left[p + q(1 + \epsilon_k)^{-2} + r(1 + \epsilon_k)^2 \right] \quad (4.3)$$

Expand the powers of $(1 + \epsilon_k)$ up to degree 3:

$$(1 + \epsilon_k)^{-2} = 1 - 2\epsilon_k + 3\epsilon_k^2 - 4\epsilon_k^3 + O(\epsilon_k^4) \quad (4.4)$$

$$(1 + \epsilon_k)^2 = 1 + 2\epsilon_k + \epsilon_k^2 \quad (4.5)$$

Let the term in brackets be $S(\epsilon_k)$:

$$\begin{aligned} S(\epsilon_k) &= p + q(1 - 2\epsilon_k + 3\epsilon_k^2 - 4\epsilon_k^3) + r(1 + 2\epsilon_k + \epsilon_k^2) + O(\epsilon_k^4) \\ &= (p + q + r) + 2(r - q)\epsilon_k + (3q + r)\epsilon_k^2 - 4q\epsilon_k^3 + O(\epsilon_k^4) \end{aligned} \quad (4.6)$$

Now compute the product $(1 + \epsilon_k)S(\epsilon_k)$:

$$\begin{aligned} 1 + \epsilon_{k+1} &= (1 + \epsilon_k) \left[(p + q + r) + 2(r - q)\epsilon_k + (3q + r)\epsilon_k^2 - 4q\epsilon_k^3 \right] + O(\epsilon_k^4) \\ &= (p + q + r) \\ &\quad + [(p + q + r) + 2(r - q)]\epsilon_k \\ &\quad + [2(r - q) + (3q + r)]\epsilon_k^2 \\ &\quad + [(3q + r) - 4q]\epsilon_k^3 + O(\epsilon_k^4) \\ &= (p + q + r) + (p - q + 3r)\epsilon_k + (q + 3r)\epsilon_k^2 + (r - q)\epsilon_k^3 + O(\epsilon_k^4) \end{aligned} \quad (4.7)$$

To maximize the order of convergence, we systematically eliminate the lower-order error terms:

1. **Consistency (Zero error condition at $\epsilon_k = 0$):**

$$p + q + r = 1 \quad (4.1)$$

2. **Quadratic Convergence (ϵ_k coefficient = 0):**

$$p - q + 3r = 0 \quad (4.2)$$

3. **Cubic Convergence (ϵ_k^2 coefficient = 0):**

$$q + 3r = 0 \implies q = -3r \quad (4.3)$$

Subtracting equation (4.2) from equation (4.1):

$$(p + q + r) - (p - q + 3r) = 1 - 0 \implies 2q - 2r = 1 \implies q - r = \frac{1}{2} \quad (4.8)$$

Substitute $q = -3r$ into this equation:

$$-3r - r = \frac{1}{2} \implies -4r = \frac{1}{2} \implies r = -\frac{1}{8} \quad (4.9)$$

Now determine q and p :

$$q = -3r = -3\left(-\frac{1}{8}\right) \implies q = \frac{3}{8} \quad (4.10)$$

$$p = 1 - q - r = 1 - \frac{3}{8} - \left(-\frac{1}{8}\right) = 1 - \frac{2}{8} \implies p = \frac{6}{8} = \frac{3}{4} \quad (4.11)$$

Investigation of Higher-Order (Can it achieve Quartic Convergence?): The coefficient of ϵ_k^3 is:

$$C_3 = r - q = -\frac{1}{8} - \frac{3}{8} = -\frac{4}{8} = -\frac{1}{2} \neq 0 \quad (4.12)$$

To make $C_3 = 0$, we would require $r = q$. Since $q = -3r$, this would imply $4r = 0 \implies r = 0, q = 0$, which yields $p = 1$ from (4.1) but $p = 0$ from (4.2), an immediate contradiction.

Thus, with 3 free parameters, the **maximum achievable rate of convergence is 3 (cubic)**.

Final Result:

$$\boxed{p = \frac{3}{4}, \quad q = \frac{3}{8}, \quad r = -\frac{1}{8}} \quad (4.13)$$

The resulting optimal iteration formula is:

$$x_{k+1} = x_k \left(\frac{3}{4} + \frac{3a}{8x_k^2} - \frac{x_k^2}{8a} \right) = \frac{1}{8}x_k \left(6 + \frac{3a}{x_k^2} - \frac{x_k^2}{a} \right) \quad (4.14)$$

with asymptotic error relation $e_{k+1} = -\frac{1}{2a}e_k^3 + O(e_k^4)$.

5 Unsolved Question 2: Optimal Parameters for Evaluating $a^{1/3}$

Problem 5.1 (Unsolved in Notes Page 3). The iterative formula:

$$x_{k+1} = px_k + \frac{qa}{x_k^2} + \frac{ra^2}{x_k^5} \quad (5.1)$$

is used for evaluating $a^{1/3}$. Find the values of p, q, r so that it converges with the **maximum rate of convergence**.

Complete Analytical Solution: Let $\alpha = a^{1/3}$ be the exact cube root, which implies:

$$\alpha^3 = a \quad \text{and} \quad \alpha^6 = a^2 \quad (5.2)$$

Define the relative error $\epsilon_k = \frac{e_k}{\alpha} = \frac{x_k - \alpha}{\alpha}$, so $x_k = \alpha(1 + \epsilon_k)$. Substituting into the iteration scheme:

$$\begin{aligned} x_{k+1} &= p\alpha(1 + \epsilon_k) + qa[\alpha(1 + \epsilon_k)]^{-2} + ra^2[\alpha(1 + \epsilon_k)]^{-5} \\ &= p\alpha(1 + \epsilon_k) + q\left(\frac{a}{\alpha^2}\right)(1 + \epsilon_k)^{-2} + r\left(\frac{a^2}{\alpha^5}\right)(1 + \epsilon_k)^{-5} \end{aligned} \quad (5.3)$$

Using $a/\alpha^2 = \alpha^3/\alpha^2 = \alpha$ and $a^2/\alpha^5 = \alpha^6/\alpha^5 = \alpha$:

$$x_{k+1} = \alpha \left[p(1 + \epsilon_k) + q(1 + \epsilon_k)^{-2} + r(1 + \epsilon_k)^{-5} \right] \quad (5.4)$$

Dividing both sides by α , where $x_{k+1}/\alpha = 1 + \epsilon_{k+1}$:

$$1 + \epsilon_{k+1} = p(1 + \epsilon_k) + q(1 + \epsilon_k)^{-2} + r(1 + \epsilon_k)^{-5} \quad (5.5)$$

Using the negative binomial expansion for $|\epsilon_k| < 1$:

$$(1 + \epsilon_k)^{-2} = 1 - 2\epsilon_k + 3\epsilon_k^2 - 4\epsilon_k^3 + 5\epsilon_k^4 - \dots \quad (5.6)$$

$$\begin{aligned} (1 + \epsilon_k)^{-5} &= 1 - 5\epsilon_k + \frac{(-5)(-6)}{2!}\epsilon_k^2 + \frac{(-5)(-6)(-7)}{3!}\epsilon_k^3 + \dots \\ &= 1 - 5\epsilon_k + 15\epsilon_k^2 - 35\epsilon_k^3 + 70\epsilon_k^4 - \dots \end{aligned} \quad (5.7)$$

Substitute these expansions into the expression for $1 + \epsilon_{k+1}$:

$$\begin{aligned} 1 + \epsilon_{k+1} &= p(1 + \epsilon_k) + q(1 - 2\epsilon_k + 3\epsilon_k^2 - 4\epsilon_k^3) + r(1 - 5\epsilon_k + 15\epsilon_k^2 - 35\epsilon_k^3) + O(\epsilon_k^4) \\ &= (p + q + r) + (p - 2q - 5r)\epsilon_k + (3q + 15r)\epsilon_k^2 + (-4q - 35r)\epsilon_k^3 + O(\epsilon_k^4) \end{aligned} \quad (5.8)$$

For the maximum rate of convergence, we set the lower-order error coefficients to zero:

1. **Order ≥ 1 (Consistency condition):**

$$p + q + r = 1 \quad (5.1)$$

2. **Order ≥ 2 (Vanishing linear error):**

$$p - 2q - 5r = 0 \quad (5.2)$$

3. **Order ≥ 3 (Vanishing quadratic error):**

$$3q + 15r = 0 \implies q + 5r = 0 \implies q = -5r \quad (5.3)$$

Substituting $q = -5r$ into equation (5.2):

$$p - 2(-5r) - 5r = 0 \implies p + 10r - 5r = 0 \implies p + 5r = 0 \implies p = -5r \quad (5.9)$$

Therefore, we have the simple relation:

$$p = q = -5r \quad (5.10)$$

Substitute $p = -5r$ and $q = -5r$ into equation (5.1):

$$(-5r) + (-5r) + r = 1 \implies -9r = 1 \implies r = -\frac{1}{9} \quad (5.11)$$

Now compute p and q :

$$p = -5 \left(-\frac{1}{9} \right) \implies p = \frac{5}{9} \quad (5.12)$$

$$q = -5 \left(-\frac{1}{9} \right) \implies q = \frac{5}{9} \quad (5.13)$$

Order and Asymptotic Error Term: Evaluate the coefficient of ϵ_k^3 :

$$C_3 = -4q - 35r = -4\left(\frac{5}{9}\right) - 35\left(-\frac{1}{9}\right) = \frac{-20 + 35}{9} = \frac{15}{9} = \frac{5}{3} \neq 0 \quad (5.14)$$

Hence, order 4 is impossible, and the maximum rate of convergence is **3 (cubic)**. The relative error relation is:

$$\epsilon_{k+1} = \frac{5}{3}\epsilon_k^3 + O(\epsilon_k^4) \quad (5.15)$$

In terms of the absolute error $e_k = \alpha\epsilon_k$:

$$e_{k+1} = \alpha \left[\frac{5}{3} \left(\frac{e_k}{\alpha} \right)^3 \right] = \frac{5}{3\alpha^2} e_k^3 + O(e_k^4) = \frac{5}{3\alpha^{2/3}} \mathbf{e_k^3} + \mathbf{O(e_k^4)} \quad (5.16)$$

Final Result:

$$\boxed{p = \frac{5}{9}, \quad q = \frac{5}{9}, \quad r = -\frac{1}{9}} \quad (5.17)$$

The resulting third-order iteration formula for $a^{1/3}$ is:

$$x_{k+1} = \frac{1}{9} \left[5x_k + \frac{5a}{x_k^2} - \frac{a^2}{x_k^5} \right] \quad (5.18)$$

6 Chebyshev-Type Second-Order Correction Method

Problem 6.1 (Notes Pages 4–5). Consider the generalized family of iterative root-finding methods:

$$x_{k+1} = x_k - \gamma_1 f(x_k) + \gamma_2 [f(x_k)]^2 \quad (6.1)$$

Determine the optimal parameters γ_1 and γ_2 to achieve a third-order (cubic) rate of convergence for finding the simple root α of $f(x) = 0$.

Derivation and Solution: Let α be the exact root so $f(\alpha) = 0$ with $f'(\alpha) \neq 0$. Let $e_k = x_k - \alpha$ and $e_{k+1} = x_{k+1} - \alpha$. Expanding $f(x_k) = f(\alpha + e_k)$ by Taylor series:

$$f(x_k) = f(\alpha) + e_k f'(\alpha) + \frac{e_k^2}{2!} f''(\alpha) + \frac{e_k^3}{3!} f'''(\alpha) + O(e_k^4) \quad (6.1)$$

Since $f(\alpha) = 0$:

$$f(x_k) = e_k f'(\alpha) + \frac{1}{2} e_k^2 f''(\alpha) + \frac{1}{6} e_k^3 f'''(\alpha) + O(e_k^4) \quad (6.2)$$

Now compute the square $[f(x_k)]^2$:

$$\begin{aligned} [f(x_k)]^2 &= \left[e_k f'(\alpha) + \frac{1}{2} e_k^2 f''(\alpha) + O(e_k^3) \right]^2 \\ &= e_k^2 [f'(\alpha)]^2 + e_k^3 f'(\alpha) f''(\alpha) + O(e_k^4) \end{aligned} \quad (6.3)$$

Subtracting α from both sides of the iteration formula (6.1):

$$e_{k+1} = e_k - \gamma_1 f(x_k) + \gamma_2 [f(x_k)]^2 \quad (6.4)$$

Substitute the series expansions of $f(x_k)$ and $[f(x_k)]^2$:

$$\begin{aligned} e_{k+1} = e_k - \gamma_1 \left[e_k f'(\alpha) + \frac{1}{2} e_k^2 f''(\alpha) + \frac{1}{6} e_k^3 f'''(\alpha) \right] \\ + \gamma_2 \left[e_k^2 [f'(\alpha)]^2 + e_k^3 f'(\alpha) f''(\alpha) \right] + O(e_k^4) \end{aligned} \quad (6.5)$$

Collecting like powers of e_k :

$$\begin{aligned} e_{k+1} = [1 - \gamma_1 f'(\alpha)] e_k \\ + \left[-\frac{1}{2} \gamma_1 f''(\alpha) + \gamma_2 [f'(\alpha)]^2 \right] e_k^2 \\ + \left[-\frac{1}{6} \gamma_1 f'''(\alpha) + \gamma_2 f'(\alpha) f''(\alpha) \right] e_k^3 + O(e_k^4) \end{aligned} \quad (6.6)$$

For the method to have **cubic convergence**, the coefficients of e_k and e_k^2 must vanish simultaneously:

1. **First-order cancellation:**

$$1 - \gamma_1 f'(\alpha) = 0 \implies \boxed{\gamma_1 = \frac{1}{f'(\alpha)}} \quad (6.7)$$

2. **Second-order cancellation:**

$$-\frac{1}{2} \gamma_1 f''(\alpha) + \gamma_2 [f'(\alpha)]^2 = 0 \implies \gamma_2 [f'(\alpha)]^2 = \frac{1}{2} \gamma_1 f''(\alpha) \quad (6.8)$$

Substitute $\gamma_1 = \frac{1}{f'(\alpha)}$:

$$\gamma_2 [f'(\alpha)]^2 = \frac{f''(\alpha)}{2f'(\alpha)} \implies \boxed{\gamma_2 = \frac{f''(\alpha)}{2[f'(\alpha)]^3}} \quad (6.9)$$

Chebyshev's Practical Formulation: In practical computation, the exact root α is unknown. Replacing the evaluation at α with evaluation at the current iterate x_k gives:

$$\gamma_1(x_k) = \frac{1}{f'(x_k)}, \quad \gamma_2(x_k) = \frac{f''(x_k)}{2[f'(x_k)]^3} \quad (6.10)$$

which yields the classical **Chebyshev Third-Order Iterative Method**:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \frac{\mathbf{f}(\mathbf{x}_k)}{\mathbf{f}'(\mathbf{x}_k)} - \frac{1}{2} \frac{\mathbf{f}''(\mathbf{x}_k)}{[\mathbf{f}'(\mathbf{x}_k)]^3} [\mathbf{f}(\mathbf{x}_k)]^2 \quad (6.11)$$

Notice that the first two terms represent the standard Newton-Raphson step, while the third term provides a curvature correction that elevates the convergence order from quadratic to cubic.

7 Unsolved Question 3: Parameter Tuning for $x_{k+1} = \frac{1}{\alpha+1}[\alpha x_k + x_k^{-2} + 1]$

Problem 7.1 (Unsolved in Notes Page 5). Given the iterative formula:

$$x_{k+1} = \frac{1}{\alpha + 1} \left[\alpha x_k + \frac{1}{x_k^2} + 1 \right] \quad (7.1)$$

- (a) Determine the algebraic equation $f(x) = 0$ whose root ξ is computed by this iteration.
- (b) Find the condition on the parameter α under which the iteration converges to ξ .
- (c) Determine the optimal value of α for the **fastest rate of convergence** (order 2), and compute the leading error constant.

Complete Analytical Solution:

Part (a): Determining the Underlying Equation

Let $\xi = \lim_{k \rightarrow \infty} x_k$ denote the fixed point of the iterative formula. Taking the limit on both sides:

$$\xi = \frac{1}{\alpha + 1} \left[\alpha \xi + \frac{1}{\xi^2} + 1 \right] \quad (7.2)$$

Multiplying both sides by $(\alpha + 1)$:

$$(\alpha + 1)\xi = \alpha \xi + \frac{1}{\xi^2} + 1 \quad (7.3)$$

Expanding the left side:

$$\alpha \xi + \xi = \alpha \xi + \frac{1}{\xi^2} + 1 \implies \xi = \frac{1}{\xi^2} + 1 \quad (7.4)$$

Multiplying through by ξ^2 (assuming $\xi \neq 0$):

$$\xi^3 = 1 + \xi^2 \iff \xi^3 - \xi^2 - 1 = 0 \quad (7.5)$$

Thus, the iteration computes the real root of:

$$\boxed{f(x) = x^3 - x^2 - 1 = 0} \quad (7.6)$$

Numerical note: Since $f(1) = -1 < 0$ and $f(2) = 3 > 0$, by the Intermediate Value Theorem, a unique positive real root exists in $(1, 2)$. Using high-precision root-finding, $\xi \approx 1.46557123$.

Part (b): Condition for Convergence

The iteration function is:

$$\phi(x) = \frac{1}{\alpha + 1} \left(\alpha x + x^{-2} + 1 \right) \quad (7.7)$$

Differentiating $\phi(x)$ with respect to x :

$$\phi'(x) = \frac{1}{\alpha + 1} \left(\alpha - \frac{2}{x^3} \right) \quad (7.8)$$

At the fixed point ξ :

$$\phi'(\xi) = \frac{\alpha - \frac{2}{\xi^3}}{\alpha + 1} \quad (7.9)$$

By the Fixed-Point Contraction Theorem, the iteration converges locally if:

$$|\phi'(\xi)| < 1 \iff -1 < \frac{\alpha - \frac{2}{\xi^3}}{\alpha + 1} < 1 \quad (7.10)$$

Assuming $\alpha + 1 > 0$ (i.e., $\alpha > -1$):

$$-(\alpha + 1) < \alpha - \frac{2}{\xi^3} < \alpha + 1 \quad (7.11)$$

- **Right inequality:** $\alpha - \frac{2}{\xi^3} < \alpha + 1 \iff -\frac{2}{\xi^3} < 1$, which is unconditionally satisfied since $\xi > 0$.
- **Left inequality:** $-(\alpha + 1) < \alpha - \frac{2}{\xi^3} \iff 2\alpha > \frac{2}{\xi^3} - 1 \iff \alpha > \frac{1}{\xi^3} - \frac{1}{2}$.

Since $\xi^3 = \xi^2 + 1 \approx (1.46557)^2 + 1 \approx 3.1479$:

$$\alpha > \frac{1}{3.1479} - 0.5 \approx 0.3177 - 0.5 = -0.1823 \quad (7.12)$$

Hence, convergence is guaranteed for:

$$\boxed{\alpha > \frac{1}{\xi^3} - \frac{1}{2} \approx -0.1823} \quad (7.13)$$

Part (c): Optimal Parameter for Fastest (Quadratic) Convergence

The rate of convergence is fastest when the first-order error term vanishes, which requires $\phi'(\xi) = 0$:

$$\phi'(\xi) = \frac{\alpha - \frac{2}{\xi^3}}{\alpha + 1} = 0 \iff \alpha - \frac{2}{\xi^3} = 0 \quad (7.14)$$

Since $\xi^3 = \xi^2 + 1$:

$$\boxed{\alpha_{\text{opt}} = \frac{2}{\xi^3} = \frac{2}{\xi^2 + 1} \approx 0.635344} \quad (7.15)$$

With $\alpha = \alpha_{\text{opt}}$, the iteration achieves **quadratic convergence (order 2)**.

Asymptotic Error Constant: Evaluating the second derivative:

$$\phi''(x) = \frac{1}{\alpha + 1} \left(\frac{6}{x^4} \right) \implies \phi''(\xi) = \frac{6}{(\alpha_{\text{opt}} + 1)\xi^4} \quad (7.16)$$

Thus, the quadratic error equation is:

$$e_{k+1} = \frac{\phi''(\xi)}{2!} e_k^2 + O(e_k^3) = \frac{3}{(\alpha_{\text{opt}} + 1)\xi^4} e_k^2 + O(e_k^3) \quad (7.17)$$

Substituting numerical values ($\xi \approx 1.46557$, $\xi^4 \approx 4.6135$, $\alpha_{\text{opt}} \approx 0.63534$):

$$e_{k+1} \approx \frac{3}{(1.63534)(4.6135)} e_k^2 \approx \mathbf{0.3976} \mathbf{e_k^2} \quad (7.18)$$

8 Unsolved Question 4: Optimal Integer k for Root of $x^3 - 5x^2 + 4x - 3 = 0$

Problem 8.1 (Unsolved in Notes Page 6). The equation $x^3 - 5x^2 + 4x - 3 = 0$ has a root near $x = 4$, which is computed by the iterative formula:

$$x_{n+1} = \frac{3 + (k-4)x_n + 5x_n^2 - x_n^3}{k}, \quad k \in \mathbb{Z} \quad (8.1)$$

Determine the value of the integer k for the **fastest rate of convergence**.

Complete Analytical Solution: Let $f(x) = x^3 - 5x^2 + 4x - 3$. Notice that the numerator can be rearranged by isolating kx_n :

$$\begin{aligned} x_{n+1} &= \frac{kx_n - 4x_n + 3 + 5x_n^2 - x_n^3}{k} \\ &= \frac{kx_n - (x_n^3 - 5x_n^2 + 4x_n - 3)}{k} \\ &= x_n - \frac{f(x_n)}{k} \end{aligned} \quad (8.2)$$

This represents a **relaxed chord / constant-slope Newton iteration**:

$$\phi(x) = x - \frac{f(x)}{k} \quad (8.3)$$

Differentiating $\phi(x)$:

$$\phi'(x) = 1 - \frac{f'(x)}{k} = \frac{(k-4) + 10x - 3x^2}{k} \quad (8.4)$$

where $f'(x) = 3x^2 - 10x + 4$.

Convergence Condition: Let ξ denote the exact root. The error equation is:

$$e_{n+1} = \phi'(\xi)e_n + O(e_n^2) = \left(1 - \frac{f'(\xi)}{k}\right)e_n + O(e_n^2) \quad (8.5)$$

For convergence, we require:

$$|\phi'(\xi)| < 1 \iff -1 < 1 - \frac{f'(\xi)}{k} < 1 \iff 0 < \frac{f'(\xi)}{k} < 2 \iff k > \frac{f'(\xi)}{2} \quad (8.6)$$

Criterion for Fastest Convergence: The convergence is fastest when the asymptotic contraction factor $|\phi'(\xi)|$ is as small as possible. The theoretical optimum occurs when:

$$\phi'(\xi) = 0 \iff 1 - \frac{f'(\xi)}{k} = 0 \iff k = f'(\xi) \quad (8.7)$$

Determination of k : We examine the problem from both standard examination perspectives:

1. **Approach 1 (Local Estimation at the Given Initial Point $x_0 = 4$):**

When the exact root is not pre-computed and the user relies on the given approximation $x \approx 4$:

$$f'(4) = 3(4^2) - 10(4) + 4 = 48 - 40 + 4 = 12 \quad (8.8)$$

Setting $k = f'(4)$ yields:

$$\mathbf{k = 12} \quad (8.9)$$

At $x = 4$, $\phi'(4) = 1 - \frac{12}{12} = 0$.

2. Approach 2 (Rigorous Asymptotic Evaluation at the Exact Root ξ):

Let us evaluate the exact root ξ near $x = 4$:

$$f(4) = 64 - 80 + 16 - 3 = -3 \quad (8.10)$$

$$f'(4) = 12 \implies x_1 = 4 - \frac{-3}{12} = 4.25 \quad (8.11)$$

$$f(4.25) = (4.25)^3 - 5(4.25)^2 + 4(4.25) - 3 = 0.453125 \quad (8.12)$$

$$f'(4.25) = 3(4.25)^2 - 10(4.25) + 4 = 15.6875 \quad (8.13)$$

$$x_2 = 4.25 - \frac{0.453125}{15.6875} \approx 4.2211 \quad (8.14)$$

The exact root to 6 decimal places is $\xi \approx \mathbf{4.220693}$. Now evaluate the exact derivative at ξ :

$$f'(\xi) = 3(4.220693)^2 - 10(4.220693) + 4 = \mathbf{15.2358} \quad (8.15)$$

To minimize $|\phi'(\xi)| = \left|1 - \frac{15.2358}{k}\right|$ over all integers $k \in \mathbb{Z}$:

- For $k = 12$: $|\phi'(\xi)| = \left|1 - \frac{15.2358}{12}\right| = |1 - 1.26965| = \mathbf{0.2697}$
- For $k = 14$: $|\phi'(\xi)| = \left|1 - \frac{15.2358}{14}\right| = |1 - 1.08827| = \mathbf{0.0883}$
- For $k = 15$: $|\phi'(\xi)| = \left|1 - \frac{15.2358}{15}\right| = |1 - 1.01572| = \mathbf{0.0157}$
- For $k = 16$: $|\phi'(\xi)| = \left|1 - \frac{15.2358}{16}\right| = |1 - 0.95224| = \mathbf{0.0478}$

Comparing the magnitudes:

$$|\phi'(\xi)|_{k=15} = 0.0157 < 0.0478 < 0.0883 < 0.2697 \quad (8.16)$$

Therefore, the globally optimal integer is:

$$\mathbf{k = 15} \quad (8.17)$$

Numerical Demonstration of Convergence:

Starting with $x_0 = 4$, we compare successive iterates for $k = 12$ vs. $k = 15$:

Iteration n	x_n with $k = 12$	x_n with $k = 15$ (Optimal)	Exact Error ($k = 15$)
0	4.00000000	4.00000000	-2.207×10^{-1}
1	4.25000000	4.20000000	-2.069×10^{-2}
2	4.21223958	4.22097778	$+2.850 \times 10^{-4}$
3	4.22271830	4.22068864	-4.178×10^{-6}
4	4.22018898	4.22069288	$+6.500 \times 10^{-8}$

Conclusion:

- If evaluated using the given starting point $x \approx 4$, the optimal integer is **k = 12**.
- If evaluated for asymptotic optimality at the exact root $\xi \approx 4.2207$, the optimal integer is **k = 15**, achieving an astonishing error reduction factor of ≈ 0.0157 per iteration.

9 Master Summary and Quick Reference Table

Iteration Method / Goal	Formula x_{k+1}	Optimal Parameters	Order	Asymptotic Error Relation
Newton-Raphson	$x_k - \frac{f(x_k)}{f'(x_k)}$	—	2	$e_{k+1} \approx \frac{f''(\alpha)}{2f'(\alpha)} e_k^2$
Square Root $\sqrt{a}$	$x_k \left(p + \frac{qa}{x_k^2} + \frac{rx_k^2}{a} \right)$	$p = \frac{3}{4}, q = \frac{3}{8}, r = -\frac{1}{8}$	3	$e_{k+1} \approx -\frac{1}{2a} e_k^3$
Cube Root $a^{1/3}$	$px_k + \frac{qa}{x_k^2} + \frac{ra^2}{x_k^5}$	$p = \frac{5}{9}, q = \frac{5}{9}, r = -\frac{1}{9}$	3	$e_{k+1} \approx \frac{5}{3a^{2/3}} e_k^3$
Chebyshev Correction	$x_k - \gamma_1 f + \gamma_2 f^2$	$\gamma_1 = \frac{1}{f'}, \gamma_2 = \frac{f''}{2(f')^3}$	3	$e_{k+1} \approx O(e_k^3)$
Cubic Eq. Root	$\frac{1}{\alpha+1} \left[\alpha x_k + \frac{1}{x_k^2} + 1 \right]$	$\alpha = \frac{2}{\xi^3} \approx 0.6353$	2	$e_{k+1} \approx \frac{3}{(\alpha+1)\xi^4} e_k^2$
Relaxed Newton ($x \approx 4$)	$x_k - \frac{x_k^3 - 5x_k^2 + 4x_k - 3}{k}$	$k = 15$ (or $k = 12$)	1	$ e_{n+1} \approx 0.0157 e_n $